

## 12 — Liquid Track: Validated Numbers, Errata, and the Gating Test Plan

### v1.4 — the quantitative record at DP-A

Two independent first-principles validation passes plus a cross-track resolution pass were run. Everything below is the surviving, corrected set — with the error trail kept deliberately visible.

#### 1. Validated quantities (DP-A: 32 °C / 80%, $\omega$ 24.2 g/kg, sink ~29 °C)

| Quantity                                    | Value                                                                       | Derivation                                                                                                                                         |
|---------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Ambient $\omega$                            | <b>24.2 g/kg</b>                                                            | Psat 4.76 kPa → Pv 3.81 → 0.622·3.81/97.5                                                                                                          |
| Target $\omega$ , 29 °C / 55%               | <b>13.8 g/kg</b>                                                            | Psat 4.01 → Pv 2.20                                                                                                                                |
| Brine floor, base 40 wt% (aw 0.45)          | <b>11.9 g/kg at 30 °C brine</b> (10.0 @27 / 14.2 @33)                       | temperature-dependent band                                                                                                                         |
| Brine floor, hot-regen 43–44 wt% (aw 0.34)  | <b>9.0 g/kg at 30 °C</b> (7.5 @27)                                          | "                                                                                                                                                  |
| Fresh-air floor (4 occupants)               | <b>48 m³/h ≈ 58 kg/h</b>                                                    | 12 m³/h/person, CO <sub>2</sub> -interlocked                                                                                                       |
| ERV pre-dry ( $\epsilon$ 0.8)               | fresh 24.2 → <b>15.9 g/kg</b>                                               | $\omega_{\text{amb}} - \epsilon(\omega_{\text{amb}} - \omega_{\text{cab}})$                                                                        |
| Mixed-mode total absorber flow              | <b>~123 m³/h (≈147 kg/h)</b>                                                | hold 13.8; gains 280 g/h ÷ (13.8 – 11.9) g/kg. One quantity in two units at p 1.2 (doc 00 §2), not a range                                         |
| <b>Peak removal, 4 adults mixed-mode</b>    | <b>0.88 kg/h</b> (0.49 with $\epsilon$ 0.8 ERV... see note)                 | CV: fresh import + occupant/envelope gains. Bare-fresh 0.88; ERV'd steady removal ~0.5–0.6                                                         |
| <b>Peak regeneration heat</b>               | <b>~0.92 kW bare / ~0.6 kW ERV'd</b>                                        | removal × 2.44 MJ/kg ÷ COP. The 0.92 kW figure is at <b>COP 0.65</b> — the with-recovery-HX value two rows down, not the 0.59 no-recovery build-up |
| <b>Daily heat, occupied DP-A duty</b>       | <b>17–20 kWh bare → ~9–11 kWh with ERV(<math>\epsilon</math> 0.8) + DCV</b> | duty-scheduled                                                                                                                                     |
| CO <sub>2</sub> -battery heat (doc 00 §5)   | <b>~1.6–2.5 kWh/day</b> for 1.6 kg CO <sub>2</sub>                          | 1.0–1.3 kWh/kg incl. water co-adsorption + sensible                                                                                                |
| COP build-up (no recovery HX)               | <b>≈0.59</b>                                                                | latent 0.678 + brine sensible 0.184 + air sensible 0.131 kWh/kg, ×1.15 losses; recovery HX → ~0.65                                                 |
| Solar array (ERV'd budget)                  | <b>2.5–4.5 m²</b> flat-plate class / ~2 m² ETC top-lift with PVT preheat    | 9–11 kWh ÷ 2.5–3 kWh/m²·day                                                                                                                        |
| Diesel heater-only, full duty               | <b>~1.1–1.4 L/day</b>                                                       | ÷ 8.5 kWh/L                                                                                                                                        |
| Absorption heat into brine                  | <b>0.66 kW peak</b>                                                         | 0.88 kg/h × 2.7 MJ/kg                                                                                                                              |
| Uncooled contactor self-limits              | +5–8 K → floor 12–14+ g/kg                                                  | air removes only ~16 W/K per 48 m³/h                                                                                                               |
| Cooling leverage                            | <b>0.5–0.7 g/kg per °C</b> of brine cooling                                 | aw 0.45 band; land cool-water sinks exploit this                                                                                                   |
| Raw-water cooling flow                      | <b>~300 L/h</b> base (2 K rise) / 600–900 heat-rich                         | 0.66 ... 2 kW rejection                                                                                                                            |
| Unattended thresholds, 1 contactor          | <b>~90 m³</b> holding 13 g/kg / <b>~160–215 m³</b> mold-safe                | 120 g/h capacity vs 0.1 ACH × $\Delta\omega$                                                                                                       |
| Overnight reserve (4 adults)                | <b>~35–40 kg</b> concentrate                                                | 12 h duty ÷ 0.143                                                                                                                                  |
| CO <sub>2</sub> at the ventilation floor    | 1,920 awake / 1,400 asleep                                                  | 0.072 m³/h ÷ 48 — <b>above the &lt;1,000 spec; closed by the doc 00 §5 stack</b>                                                                   |
| Berth cascade M-cycle                       | supply 22.6 °C · ~102 W/outlet · ~4.1 L/day                                 | working ⅓ from cabin, exhaust sat ~27 °C                                                                                                           |
| Whole-cabin once-through (X1, why deferred) | 5.5–11 kg/h absorber duty at 1–2 kW sensible                                | 370–750 m³/h all-ambient supply                                                                                                                    |
| Still driving force                         | 10–65 kPa $\Delta P_v$                                                      | pool 60–93 °C vs 29 °C-sink condenser (4–5 kPa)                                                                                                    |
| Still footprint, 0.9 kg/h                   | 0.5–0.9 m² tray                                                             | ~1–2 kg/h·m² at design $\Delta T$                                                                                                                  |
| Air-swept exhaust (degraded)                | ~98 g/kg, dew point <b>53 °C</b> → condenser recovers ~64 g/kg              | X8 rule 3; condensate technical-grade PENDING TDS                                                                                                  |
| Crystallization liquidus                    | 40 wt% ≈ <b>12–13 °C</b> ; 42 ≈ 18–19; 44 ≈ 22 °C                           | CaCl <sub>2</sub> solubility curve                                                                                                                 |
| Electrical, film primary                    | <b>25–50 W</b> initial / <b>50–80 W</b> scaled (+ERV/exhaust fan)           | fans 25–40 + recirc 15–30 + raw water 5–15 + peristaltic                                                                                           |
| Electrical, column annex                    | ~100–140 W drilled rings; 170–380 W membrane diffusers — avoid              | erratum 3                                                                                                                                          |

## 2. Errata — the correction trail (kept visible on purpose)

1. **Regeneration heat overstated 2×** in first-pass tables; corrected by re-derivation. *Lesson: never cite a margin as a computed value.*
2. **Absorber outlet is a band, not a point** (temperature-dependent); absorber cooling reclassified optional → **required**.
3. **Diffuser dynamic wet pressure was missing** from the column blower budget — membranes silently double-triple electrical draw; drilled rings win.
4. **Nylon removed** from approved materials (CaCl<sub>2</sub> ESC of polyamides).
5. **Crystallization threshold corrected** (40 wt% liquidus ~12–13 °C, not ~5); interlock restated at 42/43 wt%.
6. **Regen exhaust condensate**: 53 °C-dew-point exhaust rains salty condensate — sloped drip-leg routing mandatory; the sealed still is the logical endpoint, and the X8 condenser (doc 11 §4) is the corrected degraded mode.
7. **Sparger head constant** 1.35–1.4 kPa per 10 cm at SG 1.4.
8. **Once-through could not deliver 4-adult comfort at the governing point** (59–66% RH) — mixed-mode adopted as the occupied baseline; once-through re-scoped to unattended/low-occupancy. *Lesson: re-run every comfort claim when the design point moves.*
9. **The self-consistent steady state, not the hand chain, sizes the system** — cabin humidity, supply state, and duty must be solved together (doc 00 §4); hand-chained state tables drifted 15–30% on duty.
10. **OPEN — the ERV effectiveness behind the ERV'd duty line**. Doc 10 §3 carries an ERV pre-dried fresh state of 17.4 g/kg; §1 above carries 15.9 g/kg. Only 15.9 follows from the specified  $\epsilon \geq 0.8$  ( $24.2 - 0.8 \times 10.4$ ); 17.4 implies  $\epsilon \approx 0.65$ . The published ERV'd figures — 0.49 kg/h removal and ~0.6 kW — track 17.4, so they are effectively the  $\epsilon$  0.65 case wearing an  $\epsilon$  0.8 label; at a true  $\epsilon$  0.8 the removal falls to ~0.40 kg/h. Both values are left standing and flagged rather than reconciled on paper, because **test E measures the real  $\epsilon_{\text{lat}}$**  (and its salt-aerosol fouling trend) and will settle which case the design actually operates in. Nothing downstream is load-bearing on the difference: the bare-fresh 0.88 kg/h peak governs the sizing. *Lesson: an effectiveness quoted beside a duty is a claim about both — check that the duty was computed at the effectiveness printed next to it.*

## 3. Dominant sensitivities

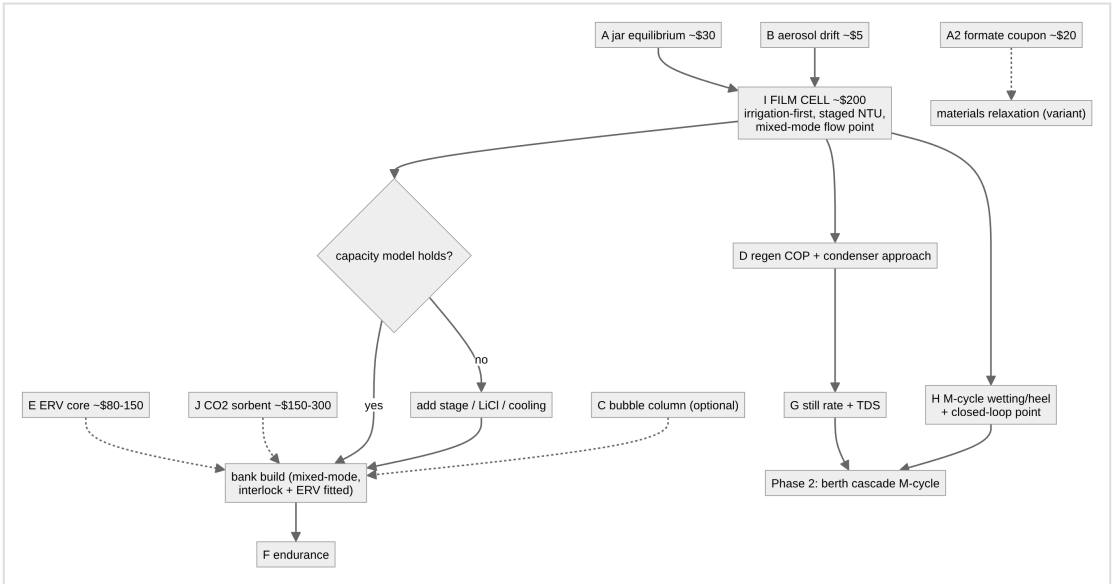
| Parameter                                                              | Uncertainty                           | Effect                                                                         |
|------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------|
| CaCl <sub>2</sub> water activity vs wt% and T                          | published data $\pm 0.05$ aw          | $\pm 1.5$ –3 g/kg on the floor — the largest unknown. <b>Measure (test A)</b>  |
| Film K·a with brine                                                    | literature 1–3 kg/m <sup>3</sup> ·s   | contactor depth 1–3 stages (test I)                                            |
| Wetting rate                                                           | 150–240 L/min·m <sup>2</sup> demanded | underwetting collapses K·a nonlinearly — test I varies irrigation <i>first</i> |
| ERV real $\epsilon_{\text{lat}}$ + fouling                             | vendor data vs salt aerosol           | gates the 9–11 kWh/day budget (test E)                                         |
| CO <sub>2</sub> -sorbent working capacity / water co-adsorption / slip | DAC literature vs this regime         | gates X10 entirely (test J)                                                    |
| Occupant moisture generation                                           | 50–90 g/h·person                      | $\pm 30\%$ on load                                                             |
| Infiltration (unattended)                                              | 0.05–0.15 ACH                         | one vs two contactors                                                          |
| Drift-eliminator + demister effectiveness                              | unquantified for brine                | gates cabin connection (test B)                                                |
| Fan/pump efficiencies                                                  | factor ~2 at small scale              | measure real amps                                                              |

## 4. Test plan — cheapest decisive experiment first

| Test                                                                            | Cost       | Decides                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A — jar equilibrium</b> (days)                                               | ~\$30      | Real aw table: sealed jars, RH probe over CaCl <sub>2</sub> 35/38/40/42/44 wt% (+LiCl arm) at ~27 and ~33 °C. <b>Go/no-go: 40 wt% @ 33 °C <math>\leq</math> 55% ERH</b>                                                                                                                                                                                                                                                                                                 |
| <b>A2 — aerated hot coupon</b> (weeks)                                          | ~\$20      | Whether potassium-formate brine survives <i>aerated</i> 80 °C service — materials relaxation for that variant only                                                                                                                                                                                                                                                                                                                                                      |
| <b>B — aerosol drift</b> (1 week)                                               | ~\$5       | Bare mild-steel coupon downstream of eliminator + demister, max face velocity, trickle <i>and</i> flooded. Any rust = redesign before cabin connection. Also arbitrates the solid track's liquid-desiccant verdict (X3)                                                                                                                                                                                                                                                 |
| <b>E — ERV core</b> (weeks)                                                     | ~\$80–150  | Real $\epsilon_{\text{lat}}$ ; salt-aerosol fouling trend (U-tube $\Delta P$ ); condensation at DP-A inlet; <b>CO<sub>2</sub> crossover / EATR &lt;5%</b>                                                                                                                                                                                                                                                                                                               |
| <b>I — film cell prototype</b> (weeks, <b>THE GATE</b> )                        | ~\$150–250 | Outlet RH vs <b>irrigation rate</b> first; 1/2/3-stage NTU/m; face velocity incl. the <b>~123 m<sup>3</sup>/h mixed-mode point</b> ; rocking rig 5–20°; flooded run; wet/dry crystallization hygiene                                                                                                                                                                                                                                                                    |
| <b>J — CO<sub>2</sub> sorbent</b> (weeks)                                       | ~\$150–300 | Working capacity at 1,000–1,500 ppm / 45–55% RH; desorption completeness 85–95 °C in the half-cycle; water co-adsorption penalty; <b>amine/ammonia slip (breathing-air gate)</b> ; oxidative fade 10 <sup>2</sup> –10 <sup>3</sup> cycles. Gates X10; fallback = oversized ERV+DCV                                                                                                                                                                                      |
| <b>J-K — potash CO<sub>2</sub> sorbent</b> (weeks; waste-heat platform variant) | ~\$40–80   | K <sub>2</sub> CO <sub>3</sub> -on-apolar-carbon: capacity at 1,000–1,500 ppm / 45–55% RH; regeneration completeness + kinetics at 120/135/150 °C with logged purge; support screen (apolar carbon vs TiO <sub>2</sub> ) incl. an <b>alumina-deactivation control</b> ; caking/deliquescence over wet-dry cycles; <b>alkaline particulate/mist carryover (breathing-air gate)</b> ; 10 <sup>2</sup> -cycle stability. Runs only where a $\geq$ ~130 °C tap exists (X11) |

| Test                                                                                          | Cost      | Decides                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L — hot-film absorption (weeks; upgrade path X12, doc 31 §2 — waste-heat platforms only)      | ~\$60–120 | Absorption rate and approach of a 43–44 wt% film at 85/88/90 °C against ~20–25 kPa steam; drain-back-on-stop and a deliberate cool-in-place fault. Reuses the sous-vide + wet/dry-bulb stack; test A's aw table feeds the lift-ceiling inequality directly. <b>Gates the AHT</b> |
| A3 — crystallizer jar extension (days; upgrade path, doc 31 §6 — mass-limited platforms only) | ~\$10     | Seeded vs unseeded 44 wt% parked at ~25 °C: supercooling degree, phase formed, caking over ~50–100 park/dissolve cycles, redissolution rate. Shares test A's jars and instrumentation. <b>Gates the static crystallizer pot</b>                                                  |
| D — regeneration COP (days)                                                                   | ~\$50     | Still tray + air-swept at 60/70/85 °C incl. PVC-tray 60 °C point and <b>condenser approach</b>                                                                                                                                                                                   |
| G — sealed-still rate (days)                                                                  | ~\$40     | kg/h·m² vs pool temp; condensate TDS (entrainment ≈ 0); <b>extended: TDS on the air-swept condenser stream</b>                                                                                                                                                                   |
| H — M-cycle wetting/heel (days)                                                               | ~\$60     | Supply temp vs feed humidity; 5/10/15° tilt; <b>closed-loop feed point (X8: recycled dried exhaust)</b> — shared verbatim with the solid track                                                                                                                                   |
| F — endurance (months, passive)                                                               | ~\$0      | Locker/closet-dryer duty: salt creep, fouling, crystallization events, ΔP trend                                                                                                                                                                                                  |
| C — bubble column (optional)                                                                  | ~\$150    | Only if the annex is ordered; sparger DWP shootout                                                                                                                                                                                                                               |

**On cost figures.** The per-test estimates above are kept because they carry the argument that matters — these are jar-and-coupon experiments, not instrumented lab campaigns — and because they let a reader sequence the program by cost. A single aggregate headline has been **retired**: the former ~\$485–755 did not decompose from this table, its span implied a costing exercise that was never done, and a currency-and-date-specific total is the first thing to rot in a record meant to be read decades from now. Read the column as **2026 order-of-magnitude, one currency, one region**. The claim the program actually rests on is scale, not precision: **the decisive liquid-track experiments together cost of order a few hundred dollars**, and test A alone — the one that moves the largest single unknown — is about \$30.



5. Rejected CO<sub>2</sub>-removal alternatives (quantified record; see doc 00 §5)

Zeolites (water outcompetes CO<sub>2</sub> at any cabin humidity) · seawater/raw-water absorption (>6 m<sup>3</sup>/h fully equilibrated, slow hydration kinetics, re-humidifies the air) · membranes (0.1 kPa partial pressure) · algae/plants (150+ kWh/day of light for 3.4 kg/day) · liquid amines (volatile slip into breathing air) · soda lime/LiOH/KO<sub>2</sub> (~15 kg/day — emergency consumable only) · electrochemical capture (R&D watch item, ~1 kWh<sub>e</sub>/kg in principle, not procurable at scale). Added by the X11 survey: open-metal-site MOFs (Mg-MOF-74 class — water poisons and hydrolyzes the sites, ~16% of capacity retained humid; poisoned-site recovery sits beyond bus grade) · flue-gas physisorbent MOFs (CALF-20 — ~0.1–0.17 mmol/g at cabin ppm, selectivity ceiling ~40–47% RH straddling the bed placement; MUF-16 class — humid-tolerant but flue-pressure-validated, marginal at 0.1–0.15 kPa, cobalt-dust gate) · fluorinated ultramicroporous physisorbents (SIFSIX/TIFSIX/NbOFFIVE — genuinely ppm-capable at ~1.2–1.3 mmol/g, but 7.5–10 mmol/g water co-adsorption rides every swing, SIFSIX-3-Ni phase-changes at ~50% RH, humid/oxidative cycling instability, and a fluorometallate-pillar degradation path upstream of breathing air) · moisture-swing quaternary-ammonium resins (commodity AERs, but regeneration-by-wetting evaporates its water into the cabin and re-exports it onto the brine loop — order 5–15 kWh/day of displaced regeneration heat vs 1.6–2.5 for TSA, an X8 rule-1 violation at DP-A; capture also degrades at moderate-plus RH) · supported K<sub>2</sub>CO<sub>3</sub> **at solar grade** (equilibrium-dead below ~120 °C — adopted instead as the waste-heat-platform bed: X11, test J-K).

6. Open questions (logged, not load-bearing)

- Evaporative-media life in concentrated CaCl<sub>2</sub> through wet/dry cycles (inside I/F).
- LiCl sourcing vs accepting hot-regen CaCl<sub>2</sub> as the deep-dry lever; LiCl worsens wetting (revalidate irrigation margin if spiked).
- Fans/pumps: independent units vs single + manifold — measured amps settle it.
- Cool-humid (non-tropical) absorber behavior; cold-climate land installs.

- Heat-driven air pumps (NIFTE/fluidyne/thermoacoustic) — R&D track only.
- Upgrade-path watch items (doc 31, none load-bearing): AHT absorber materials at 90 °C brine (PVDF vs PP margin); MVR compressor mist carryover (test-B analogue pre-gate) and non-condensables purge cadence; coupled-HP refrigerant choice (R290 procurement vs R600a fit); hexahydrate PCM nucleator persistence (logged, low priority at DP-A).
- Distillate potability chain (baffle + TDS + carbon polish): one lab water test before regular drinking (safety register item 5).
- CO<sub>2</sub>-sorbent watch register (research-grade, no supply chain; any promotion inherits full test-J scope): **MOF-808-AA** (glycine/lysine-appended Zr-MOF — nonvolatile anchored amines, water-*enhanced* uptake at ~50% RH, mild desorption) primary; **tetraamine-Mg<sub>2</sub>(dobpdc)** (supersedes the mmen/diamine class, whose amine volatilization its own authors concede; steam-regenerable, ppm-validated) secondary; SIFSIX-18-Ni-β a literature line only (hydrophobic trace-CO<sub>2</sub> physisorbent — no supply chain, fluoride-pillar caveat).

## 7. Bottom line

---

The physics closes on independent passes and one cross-track resolution pass. The occupied claim now reads: **4-adult comfort at DP-A stands on the mixed-mode baseline with cooled base brine, ERV, and the CO<sub>2</sub> stack** — conditional on tests A/B/E/I/J, all bench-cheap. Two-adult and unattended claims stand on the base once-through system. Phase 2 (berth cascade M-cycle) adds no new load-bearing physics *in the cascade topology*; whole-cabin AC is deferred on heat-scale grounds, where the two tracks converge.

---

*Part of an open defensive-publication release: hardware CERN-OHL-P v2, text CC-BY-4.0. No patents sought or held. Unbuilt paper design — see LICENSE.*

### Version history

- **v1.0** — New lineage from archived core-03: all quantities restated at DP-A; errata 8–9 added from the resolution passes; tests E and J and the P16 extensions integrated; CO<sub>2</sub>-alternatives record added.
- **v1.1** — Test J-K added (§4); CO<sub>2</sub>-rejection record extended with the X11 survey — open-metal-site MOFs, CALF-20, MUF-16, fluorinated HUMs, moisture-swing AERs, solar-grade potash (§5); the mmen upgrade line replaced by the CO<sub>2</sub>-sorbent watch register — MOF-808-AA primary, tetraamine-Mg<sub>2</sub>(dobpdc) secondary (§6).
- **v1.2** — Upgrade-path tests L (AHT hot-film absorption, X12) and A3 (crystallizer jar extension) added to §4 as platform-conditional entries outside the baseline gating budget; doc 31 watch items added to §6.
- **v1.3** — Clarifying pass from the parameter register (doc 50 §3.5), no design figure changed: §1's absorber-flow row restated as one quantity in two units (~123 m<sup>3</sup>/h ≈ 147 kg/h) rather than a range, and the COP behind the 0.92 kW peak named as the with-recovery 0.65; **erratum 10 opened** — the ε<sub>lat</sub> inconsistency behind the ERV'd duty line, left standing and gated on test E; §4 gains a budget note recording that the ~\$485–755 headline no longer decomposes from the table beneath it.
- **v1.4** — Aggregate cost headline retired from the §4 title (it did not decompose from the table and dates badly in a prior-art record); per-test estimates retained and explicitly scoped as 2026 order-of-magnitude figures.